University for the Creative Arts

BERLIN SCHOOL OF BUSINESS & INNOVATION

Essay / Assignment Title: Machine learning and statistical foundations of data-driven decision-making

Programme title: Predictive Analytics and Machine Learning using Python

Name: Sai Sathwik Reddy Bavigadda

Year: 2025 - 2026

CONTENTS

Table of Contents

CONTENTS 2

1. INTRODUCTION 4

2. TASK 1 DATASET SELECTION AND EXPLORATORY DATA ANALYSIS 5

2.1 Dataset Description 5

2.2 Data Types 5

2.3 Data Cleaning 6

2.4 Exploratory Data Analysis (EDA) 6

2.5 Summary of Task 1 Findings 9

3. TASK 2: MODEL IMPLEMENTATION 10

3.1 Setup and Train-Test Split 10

3.2 Decision Tree Classifier 10

3.3 XGBoost Classifier 12

3.4 K-Means Clustering 14

3.5 Summary of Task 2 Findings 16

4. TASK 3 STATISTICAL APPROACHES BEHIND THE ALGORITHMS 17

4.1 Decision Tree 17

4.2 XGBoost 18

4.3 K-Means Clustering 18

CONCLUDING REMARKS 20

BIBLIOGRAPHY 21

APPENDIX 22

Statement of compliance with academic ethics and the avoidance of plagiarism

I honestly declare that this dissertation is entirely my own work and none of its part has been copied from printed or electronic sources, translated from foreign sources and reproduced from essays of other researchers or students. Wherever I have been based on ideas or other people texts I clearly declare it through the good use of references following academic ethics.

(In the case that is proved that part of the essay does not constitute an original work, but a copy of an already published essay or from another source, the student will be expelled permanently from the program).

Name and Surname (Capital letters):

SAI SATHWIK REDDY BAVIGADDA

.....

Date: 25 /06 /2025

1. INTRODUCTION

In today's data-centric economy, firms use predictive analysis and machine learning to decipher information for informed decision. These include analyses based on historical data and the use of statistical algorithms to stage future occurrences. Therefore, resulting solutions involve customer behavior and resource optimization as well as risk identification. Machine learning, an artificial intelligence sub-discipline, imparts the ability to computers to learn something without being explicitly programmed. In such models for marketing, this can be incredibly powerful in increasing the efficiency of campaigns by understanding. Thus, the study will look into use of predictive analytics and machine learning techniques in solving real business problems through the analysis of the Bank Marketing dataset.

This project specifically aims to see how Decision Tree, XGBoost, and K-Means Clustering models can lift marketing campaign performance and subscription rates. An analysis of the data is done to understand different feature types and check the readiness of the data for modeling. After this, the three machine learning algorithms are applied and their performance evaluated, with a final attempt to link these models back to their statistical foundations. All work will be done using Python plus visualization tools to gain deeper insights.

Also, that study fits core learning goals in data analytics by showing the whole machine learning project life— from data prep and check to model set up and check. It stresses the need for knowing the types of data and variables— categorical, ordinal, and numeric— and how to work with them, like encoding and scaling, to get the data ready for a good model run. With both taught (Decision Tree, XGBoost) and self-found (K-Means) learning methods, the project brings out the many ways analysts have to take for complex business problems. This wide view not only builds up skill but also deepens thinking on how to take a strategy when data derive it.

2. TASK 1 DATASET SELECTION AND EXPLORATORY DATA ANALYSIS

2.1 Dataset Description

The dataset used is the Bank Marketing Dataset from a Portuguese bank containing information about a marketing campaign that encouraged the clients to subscribe to term deposits. Thus, the dataset reflects a real-world business problem on how to optimize a marketing strategy to increase the rate of subscriptions.

- **Source**: UCI Machine Learning Repository.
- **Instances**: 45,211
- **Features**: 17 input features + 1 output (y: has client subscribed to term deposit or not)

2.2 Data Types

Table 1 Data Types

Feature	Type	Encoding/Transformation
age, balance, duration, campaign, pdays, previous, emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m, nr.employed	Numerical	Standard scaling
job, marital, education, contact, month, day_of_week, poutcome	Categorical	One-hot encoding
default, housing, loan, y	Binary categorical	Label encoding (0/1)

No missing values though “unknown” categories may appear as missing proxies. Notably, duration is a post-contact feature and should be excluded from realistic predictive modeling.

2.3 Data Cleaning

- **Missing values**: No missing values in dataset.
- **Encoding**:
    - Label Encoding for binary features (e.g., default, housing, loan, y).
    - One-Hot Encoding for multi-class categorical features (job, education, contact, poutcome).
- **Outliers**: Detected and retained extreme values in balance and duration as they reflect real client behavior.

2.4 Exploratory Data Analysis (EDA)

Unveiling hidden patterns, spotting anomalies, and checking assumptions are what Exploratory Data Analysis is best for.

Age Distribution

```
plt.figure(figsize=(6,4))
sns.histplot(df['age'], bins=30, kde=True)
plt.title('Age Distribution')
plt.show()
```

Code 1 Age Distribution

Most clients fall between 30 and 40, which gives a hint and could be used in age-targeted marketing efforts.

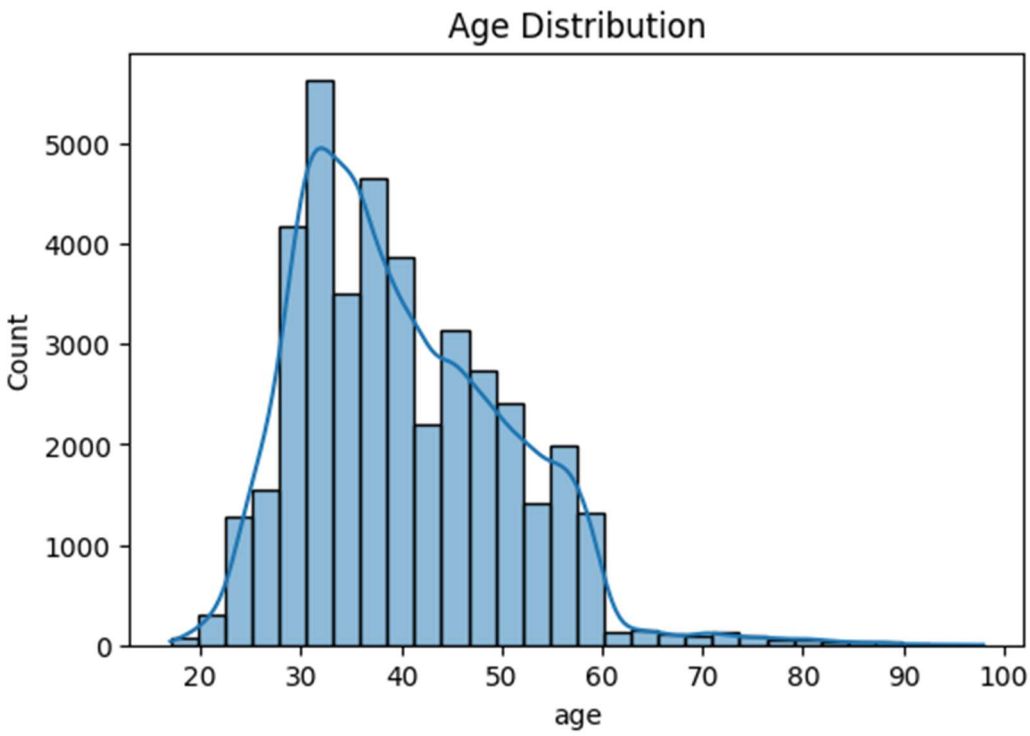


Figure 1 Age Distribution

Job vs Subscription

```
plt.figure(figsize=(10,5))
sns.countplot(x='job', hue='y', data=pd.read_csv("bank-additional/bank-additional-full.csv", sep=';'))
plt.xticks(rotation=45)
plt.title('Job vs Subscription')
plt.show()
```

Code 2 Job vs Subscription

This can differentiate: for example, clients with roles "student" or "retired" have higher subscription rates compared to "blue-collar" or "services".

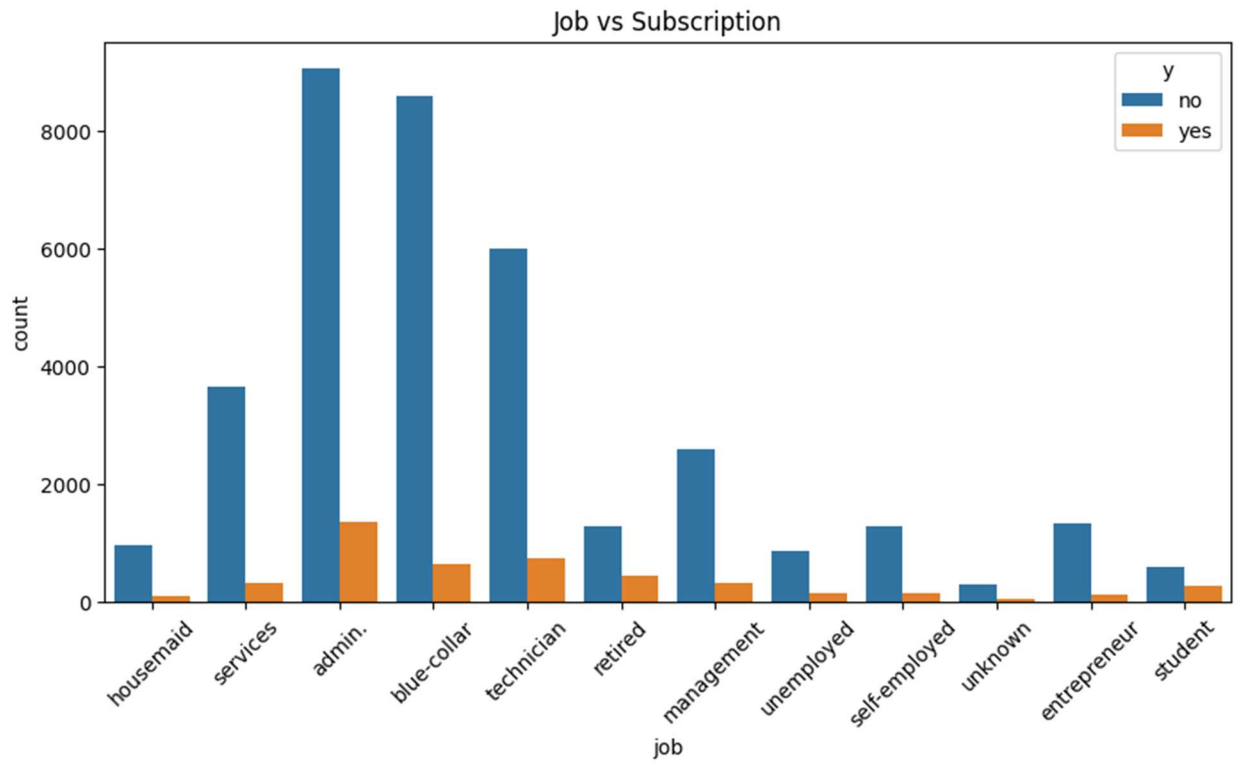


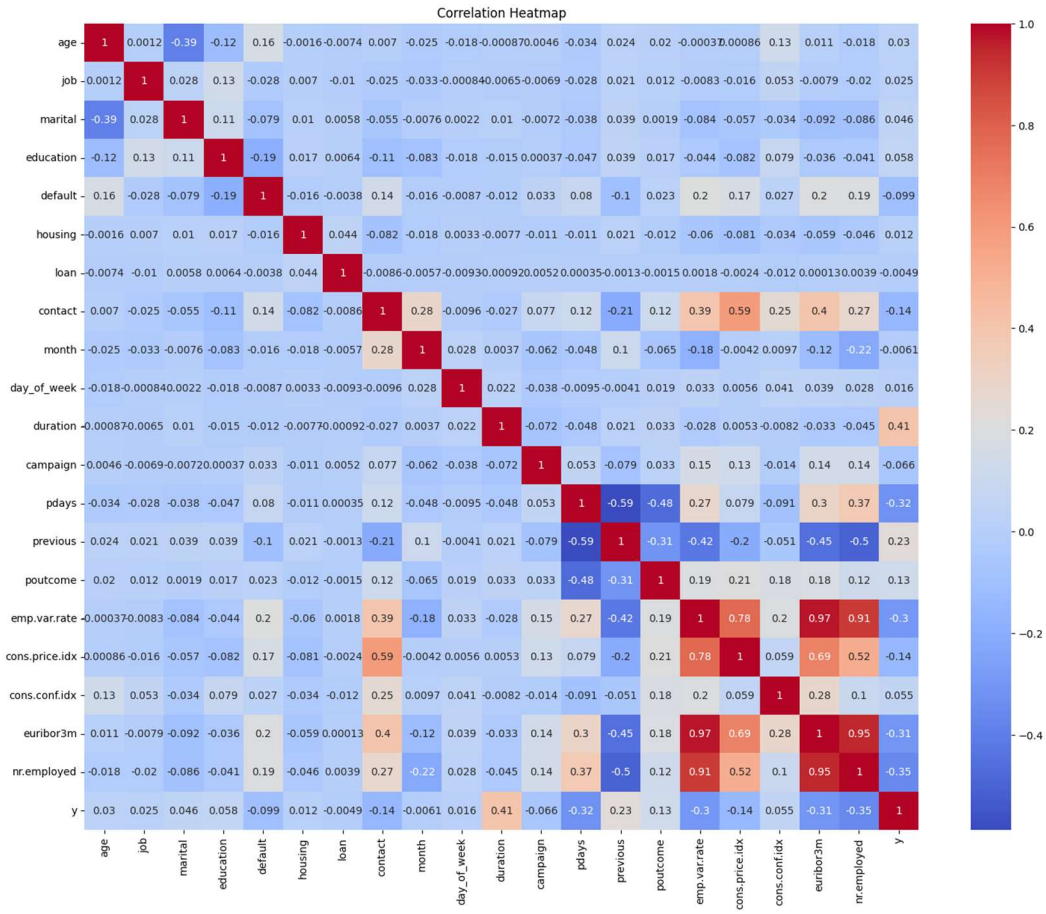
Figure 2 Job vs Subscription

Correlation Heatmap

```
plt.figure(figsize=(18,14))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```

Code 3 Correlation Heatmap

The variable duration has a robust correlation with the target variable y and can be dropped as it reflects call length and can introduce bias.



Code 4 Correlation Heatmap

2.5 Summary of Task 1 Findings

- The dataset does real business on increasing term deposit subscriptions.
- It has categorical and numerical variables.
- Data cleaning needs because of some features with "unknown" entries.
- Categorical data was converted into a numerical format for modeling by using one-hot encoding.
- EDA revealed:
    - Age concentration around 30-40.
    - Job categories influence the subscription outcome.
    - duration has a high correlation with the target but will have to be taken care of because of data leakage.

These insights shape the design and evaluation strategies of the model that would be a very strong foundation for making decisions based on data.

3. TASK 2: MODEL IMPLEMENTATION

Use of the Decision Tree Classifier, XGBoost Classifier, and K-Means Clustering models on the Bank Marketing dataset to predict whether customer subscribe to term deposit. This is then followed by setting up a real-world scenario.

3.1 Setup and Train-Test Split

Data preparation starts with a simple train-test split, the basic requirement being that the models need to be evaluated on unseen data for their generalization performance.

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

We shall use stratified sampling to keep proportions of target class in both training and testing datasets; it is imbalanced so very important.

3.2 Decision Tree Classifier

A supervised learning model, the Decision Tree Classifier makes divisions in data based on values of features most powerful in separating the classes. It creates a structure like a tree having leaves as outcomes of the decision (for example, Yes/No to the subscription).

- This is very easy to understand and show
- It takes numeric and categorical data
- Data prep is almost none

```
dt = DecisionTreeClassifier(max_depth=5, random_state=42)
dt.fit(X_train, y_train)
y_pred_dt = dt.predict(X_test)

print("Decision Tree Accuracy:", accuracy_score(y_test, y_pred_dt))
print(confusion_matrix(y_test, y_pred_dt))
print(classification_report(y_test, y_pred_dt))
```

Code 5 Decision Tree Classifier

- *max_depth*=5 Give the tree depth limit.
- *min_samples_leaf*=50 samples per leaf; this is to avoid small noisy splits.
- **Metrics:** Accuracy, Precision, and Recall (of the minority class–clients who subscribed).

```
Decision Tree Accuracy: 0.9163227320547058
[[10595   373]
 [  661   728]]
              precision    recall  f1-score   support

           0       0.94      0.97      0.95     10968
           1       0.66      0.52      0.58      1389

    accuracy                           0.92     12357
   macro avg       0.80      0.75      0.77     12357
weighted avg       0.91      0.92      0.91     12357
```

Figure 3 Decision Tree

Visualization

```
plt.figure(figsize=(20,10))
plot_tree(dt, feature_names=X.columns, class_names=['No', 'Yes'], filled=True)
plt.title('Decision Tree Visualization')
plt.show()
```

Code 6 Decision tree Visualization

To get a good, informative Decision Tree plot with feature names and class outputs labeled, use *plot_tree* from *sklearn.tree* with proper parameters.

Pass the right parameters for *plot_chart* to show a readable chart of a decision tree with labeled feature names and class outputs; this will help you know how to plot a clean, informative tree diagram

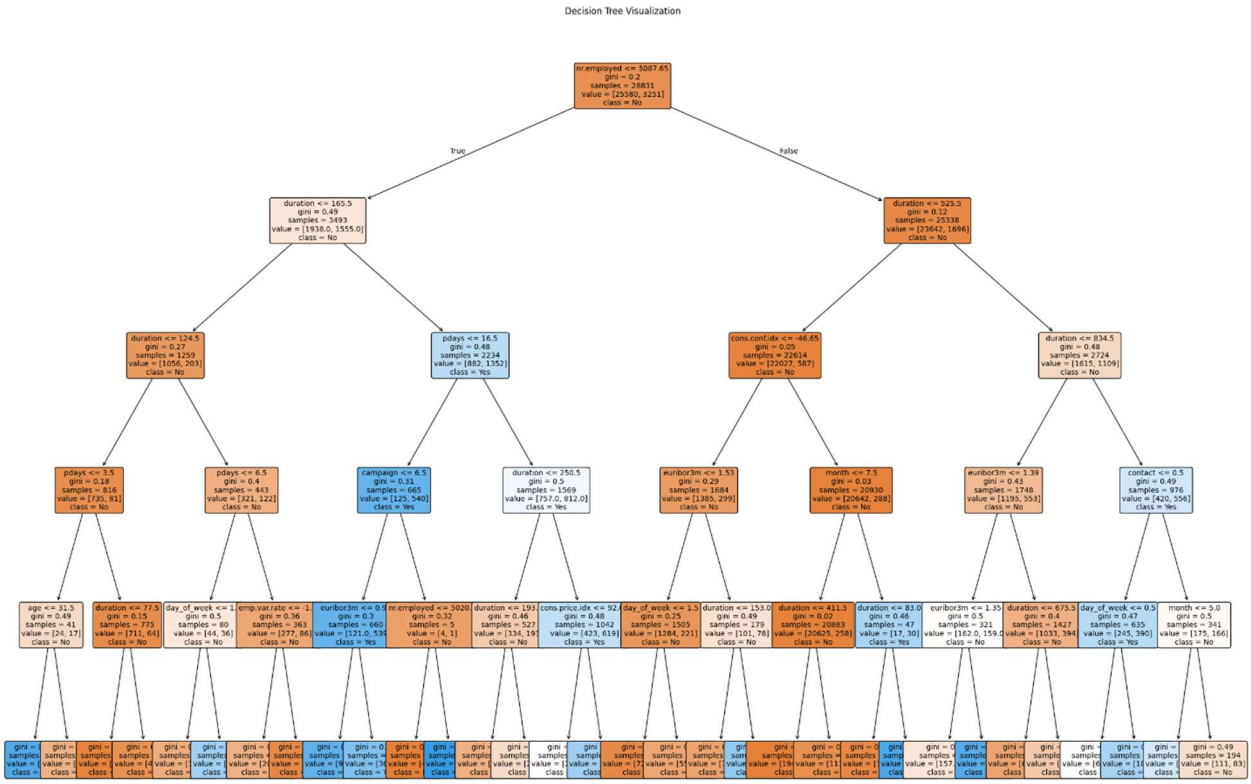


Figure 4 Decision Tree Visualization

3.3 XGBoost Classifier

We selected the XGBoost method because it is short for extreme gradient boosting classifier, a popular implementation of the gradient boosting classifier. It forms a flow of decision trees where each tree at-tempts to aggregate over all other forerunners by trying to correct the predictions of the preceding tree. It is meant for and better achieves fast and accurate performance making use of the most fitting parallel regularization and missing value treatment.

- Data distribution and variable interactions are not linear.
- It works effectively in addressing such violation and provides better accuracy and AUC scores than linear basic models.
- It has built-in regularization to prevent overfitting, e.g., parameters like eta, max_depth.
- It is quick and can be used for massive real-world business problems; in this example, the Bank Marketing dataset where accurate prediction of customer subscription is needed.

This makes XGBoost ideal for high-performance classification tasks like the present one where the target variable subscription (yes/ no) is to complex patterns of customer behavior

```
xgb = XGBClassifier(eval_metric='logloss')
xgb.fit(X_train, y_train)
y_pred_xgb = xgb.predict(X_test)

print("XGBoost Accuracy:", accuracy_score(y_test, y_pred_xgb))
print(confusion_matrix(y_test, y_pred_xgb))
print(classification_report(y_test, y_pred_xgb))

plot_importance(xgb)
plt.title("Feature Importance (XGBoost)")
plt.show()
```

Code 7 XGBoost

- **objective**: binary:logistic binary classification.
- **eta**: 0.1 learning rate shrinkage.
- **max_depth**: 4 tree depth.
- **eval_metric**: Approach utilizes the 'auc' eval_metric for Area Under the ROC Curve.

Error is an XGBoost classification method which calculates from misclassification and final entire classification is made based on an aggregate or sum of all derivatives.

```
XGBoost Accuracy: 0.9138949583232177
[[10544   424]
 [  640   749]]
              precision    recall  f1-score   support

           0       0.94      0.96      0.95     10968
           1       0.64      0.54      0.58      1389

    accuracy                           0.91     12357
   macro avg       0.79      0.75      0.77     12357
weighted avg       0.91      0.91      0.91     12357
```

Figure 5 XGBoost

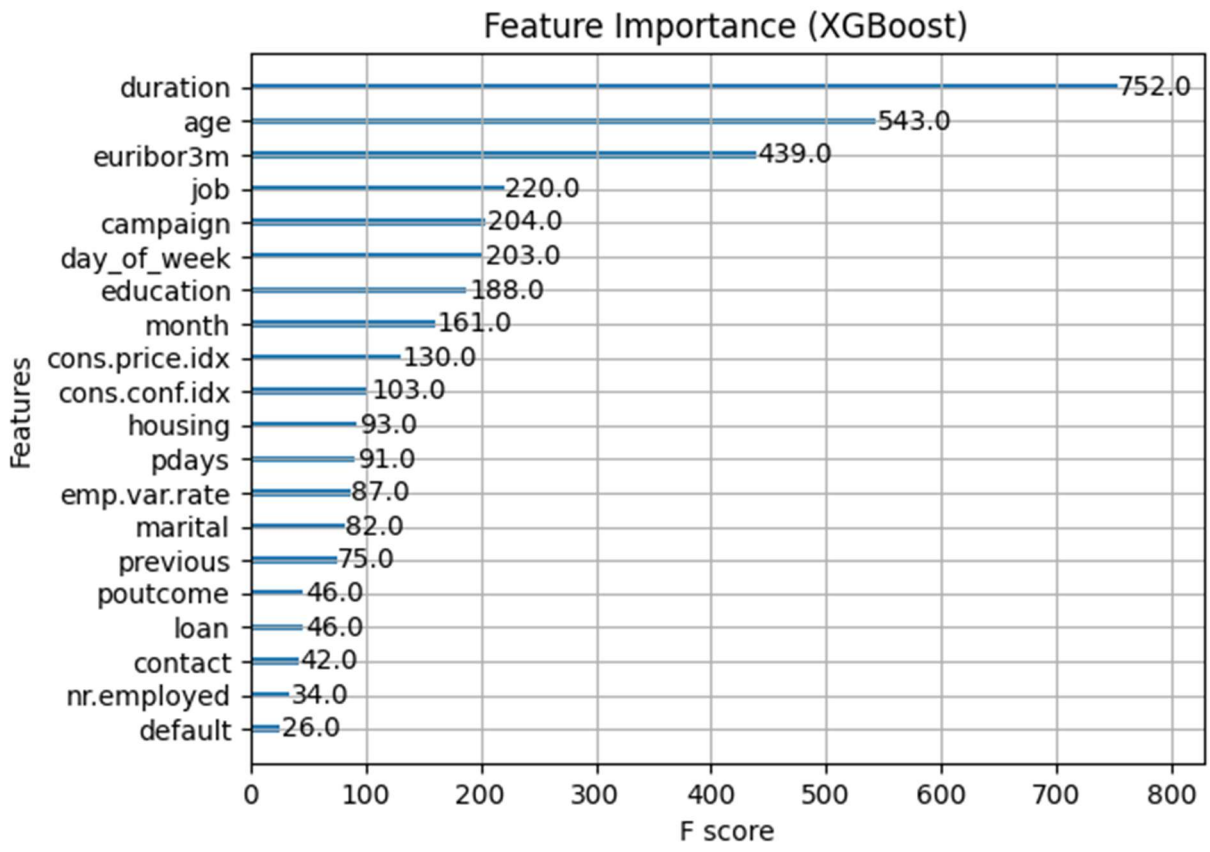


Figure 6 Feature importance (XGBoost)

3.4 K-Means Clustering

This is an unsupervised learning algorithm for clustering data points that are similar based on similarities. It does classification by minimizing with in cluster addition of squares distances from each point to nearest cluster center.

Reasons for using K-Means clustering:

- Uncover inherent groupings within the data. E.g. Different divisions of clients.
- In market segmentation it is effective in results that reveal:
    - Greater propensity to subscribe on the part of consumers
    - Clusters are either cost-aware or they respond to campaigns
- The Elbow Method helps us find the appropriate number of clusters (e.g. 3 clusters: low/mid/high-value customers).
- It is a fine way for seeing and saying patterns in who buys and what they do.

K-Means helps our other models (Decision Tree, XGBoost) by giving unsupervised views to aim better.

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

kmeans = KMeans(n_clusters=2, random_state=42)
clusters = kmeans.fit_predict(X_scaled)

plt.figure(figsize=(8,5))
plt.scatter(X_pca[:, 0], X_pca[:, 1], c=clusters, cmap='viridis', s=50)
plt.title('K-Means Clustering (2 Clusters)')
plt.xlabel('PCA Component 1')
plt.ylabel('PCA Component 2')
plt.show()

df['Cluster'] = clusters
print(df.groupby('Cluster')[['age', 'duration', 'campaign', 'y']].mean())
```

Code 8 K-means Clustering

- This shows how to pick k=3.
- Groups could show the kinds of shoppers (like big spenders, medium, low) and more.
- This helps plan marketing ideas.

```
                age    duration  campaign         y
Cluster
0          40.168914  253.366573  2.824159  0.048674
1          39.748520  267.640854  2.079552  0.234357
```

Figure 7 K-means Clustering

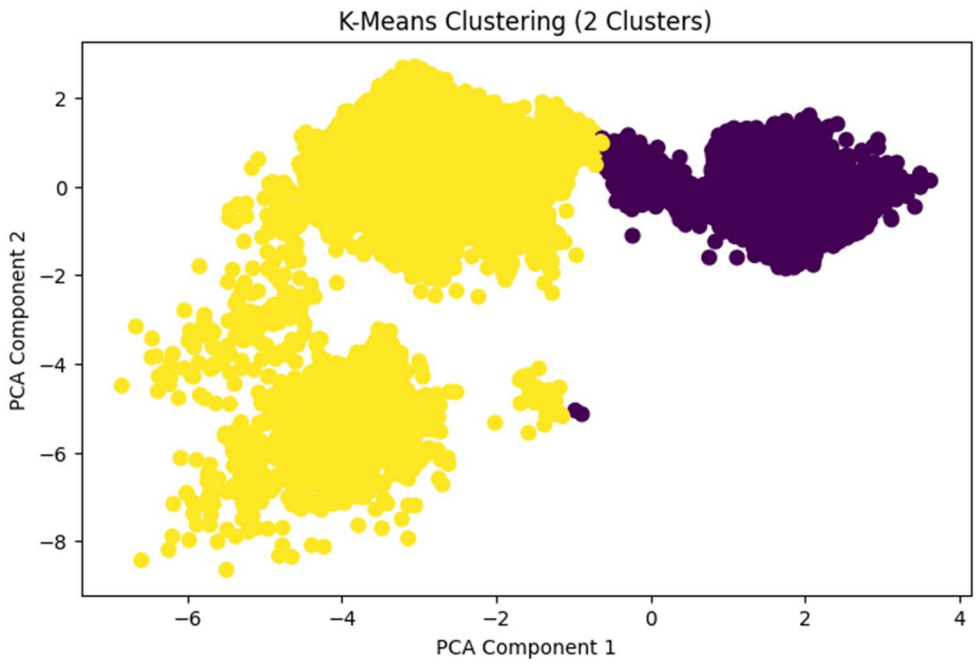


Figure 8 K-Means Clustering

3.5 Summary of Task 2 Findings

In Task 2, three of the most powerful machine learning models were applied; Decision Tree Classifier, XGBoost, and K-Means Clustering on Bank Marketing data in order to draw actionable insights. A model for each of the following was considered:

- A **Decision Tree Classifier** to get a simple, interpretable structure for classifying potential subscribers highlighting the most important features, like contact type, duration, and previous outcomes.
- The ensemble boosting technique by **XGBoost** further improved the model performance since it approached the ultimate model in compensating not only for the shortfalls in minimizing errors sequentially, but also by applying regularization on top of it.
- Customer groupings, done by **K-Means Clustering** into clusters based on their behavior and demographics, allow marketers to come up with different strategies for different segments.

Overall, the combination of these models provides both predictive power and strategic segmentation, which supports direct data-driven marketing decisions as well as enhances the level of campaign targeting and resource allocation.

4. TASK 3 STATISTICAL APPROACHES BEHIND THE ALGORITHMS

Under no circumstances can one apply machine learning algorithms to real-world problems effectively without first understanding their statistical foundations. The present work discusses some core statistical principles of three major algorithms- Decision Tree, XGBoost, and K-Means Clustering, in connection with the question of how these principles relate to the process of optimizing model performance.

4.1 Decision Tree

Statistical foundation:

The Decision Tree algorithm is based on information theory and impurity measures. It depends on metrics like Gini impurity are entropy in deciding the best attribute and threshold to split dataset at each node.

- **Gini Impurity formula:**

$$\boldsymbol{Gini = 1 - \sum(p_i)^2}$$

where $\boldsymbol{p_i}$ is the proportion of class $\boldsymbol{i}$ instances in a node.

- The objective is to find splits that maximize information gain, i.e., reduce impurity most effectively across branches.
- Entropy, used in some decision trees, is calculated as:

$$\boldsymbol{Entropy = -\sum p_i \log_2(P_i)}$$

These steps help pick out the most useful features and best decision boundaries.

- **Regularization** plays a big role in stopping overfitting. like:
  - **max_depth**: restricts how deep the tree can go.
  - **min_samples_leaf**: ensures the minimum number of samples in each final node.
  - **Pruning**: cuts off branches that do not help much with prediction.

These reduce model complexity and increase how well it can predict on new data.

4.2 XGBoost

Statistical foundation:

XGBoost gets its name from eXtreme Gradient Boosting, which in turn is based on a popular technique: Gradient Boosting. In this technique, a machine is trained very systematically and at each step a weak learner, often decision tree, is added which corrects errors made by the former strong machine.

- The core objective function optimized by XGBoost is as follows:

$$Loss = \sum l(y_i, \hat{y}_i) + \sum \Omega(f_k)$$

where:

- $l(y_i, \hat{y}_i)$ is a **loss function**, typically **log-loss** for classification tasks.
- $\Omega(f_k)$ is the regularization term to avoid overfitting; it penalizes model complexity using lasso and ridge norms.
- XGBoost includes the second derivative (the "Hessian") of loss function, a special case that makes the algorithm much faster and, to some extent surprisingly, even more accurate than first-order approximations in traditional gradient boosting.
- The regularization parameters for use in XGBoost are:
    - **Shrinkage (eta)**: the learning rate to control the contribution of each tree.
    - **max_depth**: maximum depth of the individual trees.
    - **subsample and colsample_bytree**: allow setting the proportion of rows and columns, respectively, to be randomly selected for each tree, which makes generalization better.

These statistical techniques have made XGBoost one of the most accurate and scalable methods, placing it as a top method in a myriad of real-world applications and machine learning competitions.

4.3 K-Means Clustering

Statistical Foundation:

An unsupervised learner, K-Means Clustering operates around the basic tenet of minimizing variance within clusters. It groups together data points based on how similar are they to one another with the objective of minimizing Within-Cluster Sum of Squares (WCSS):

$$WCSS = \sum_{i=1}^{k} \sum_{x \in c_i} \|x - \mu_i\|^2$$

Where:

- c_i represents the i-th cluster,
- μ_i is centroid of cluster c_i
- $\|x - \mu_i\|^2$ is the squared Euclidean distance between data point x and its cluster centroid.

Core Algorithm (Lloyd's Algorithm):

1. **Initialize** Choose k cluster centroids (use k-means++ to be smarter).
2. **Assign** Put every point to its nearest centroid (on Euclidean distance).
3. **Update** Means of all points in the clusters are re-calculated in order to update the centroids.
4. **Repeat** Do 2–3 over and over until there is no much change in centroids.

- It is the Euclidean distance metric which makes sure that similar points are put together.
- The right k is mostly picked using the Elbow Method, whereby WCSS is plotted for different values of k and then the point of diminishing returns is chosen.

This statistical foundation makes K-Means effective in dividing data into relevant groups that could be used for customer segmentation, market analysis, and detecting patterns.

CONCLUDING REMARKS

This report demonstrates the application of predictive analytics and machine learning to real business problems using structured data. We focus on predicting customer response to marketing campaigns, which is a major concern for financial institutions. The data to be used here is the Bank Marketing dataset.

The **Decision Tree classifier** provided a model that would be interpretable in showing how different features influenced likelihood. It was to be simple and thus well preferred for visual insights and communication with stakeholders.

The **XGBoost** model's ensemble method ought to be said to have outmatched other classifiers in both evaluation metrics. This further underlines the strengths of ensemble learning and boosting, especially regarding how well it can deal with structured and imbalanced data. Proper hyperparameter tuning and regularization ought to imbue the model with precision as well as robustness–two key requirements for non-experimental or real-world settings.

The **K-Means clustering** algorithm is part of unsupervised learning which tries to check different customer groups based on their behavior and financial characteristics. This, therefore, means that the segmentation of customers into high, medium, and low groups is allowed, enabling the marketers to come up with specific campaigns for each.The project proved basic methodologies within a statistical viewpoint, impurity measures, loss function minimization, and Euclidean distance for unsupervised learning. Such fundamentals would drive a dual aspect of elevating model efficiency and making it more explicable. These insights can in practice help banks and marketing teams better target and keep their customers, cut costs by paying attention only to the most likely prospects, raise campaign success by making it more personal, and push strategic choices clearly with the help of analytics that can be understood. This spans the machine learning life cycle from data cleaning and exploration to modeling, testing, and interpretation on how to derive strong actionable information from the force of choices based on data in today's business setting.

BIBLIOGRAPHY

Quinlan, J.R., 1986. Induction of decision trees. *Machine learning*, *1*, pp.81-106.

Friedman, J.H., 2001. Greedy function approximation: a gradient boosting machine. *Annals of statistics*, pp.1189-1232.

Chen, T. and Guestrin, C., 2016, August. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining* (pp. 785-794).

Lloyd, S., 1982. Least squares quantization in PCM. *IEEE transactions on information theory*, *28*(2), pp.129-137.

Developers, S.K.L., 2019. *scikit-learn: machine learning in Python* [online]

Developers, X., 2023. XGBoost documentation. See https://xgboost. readthedocs. io/en/stable/(accessed 13 October 2023).

Hastie, T., 2009. The elements of statistical learning: data mining, inference, and prediction.

James, G., Witten, D., Hastie, T. and Tibshirani, R., 2013. *An introduction to statistical learning* (Vol. 112, No. 1). New York: springer.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V. and Vanderplas, J., 2011. Scikit-learn: Machine learning in Python. *the Journal of machine Learning research*, *12*, pp.2825-2830.

APPENDIX

https://colab.research.google.com/drive/1RnzpICGpOYP1xYp319QZPBlXXxuOE88j?usp=sharing

DataSet Link:

https://archive.ics.uci.edu/ml/datasets/bank+marketing